

EMBA756 MANAGING OPERATIONS

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BOOK REPORT ON THE "GOAL "

What is **The Goal**? The book claims to be many things. It claims to be about science and education, about progress, about global principles of manufacturing and some of you might say that it is about a love story (for those of you that think so, please let me know your names so I can preassign your course grades). For me the book is

- (a) a fascinating description of an operations process (for those of you without an operations background, an excellent opportunity to get an introduction to manufacturing environments),
- (b) an excellent example of how operations should be viewed: as a **business function** which can be understood with the use of a **business language** and not with the use of obscure technical terminology, and
- (c) a powerful demonstration that the important, and in most cases only, prerequisite for becoming an operations manager is **common sense** and **intellectual attraction** to environments that exhibit **complexity** and **variability**.

I recommend that you treat the reading of **The Goal** as a leisure activity and pace it over a few days. The danger is that some of you might find it such an irresistible reading that you might want to finish it overnight. (I am not responsible for any side effects of your obsessed behavior with this book, such as family problems and/or your lover(s) deciding to leave you). After you finish reading the book, answer the following questions:

1. Give me the definitions of throughput, inventory and operational expense given in The Goal. How do they compare with the traditional definitions? Do you find them useful, and why?
2. Give me the definition of a bottleneck operation. Develop your own simple example to demonstrate it to me. Describe two or three different ways to discover the bottleneck of a process. Is it true that the “bottleneck” of a process is the “slowest operator or equipment” in it? Can efficient operators or processes be the bottlenecks of a multi-stage workflow process? Why or Why not?
3. Draw a top-level process flow diagram for Alex’s factory. You may wish to first focus on the NCX-10 and then aggregate the rest of the machines into three categories of process steps: those that precede the NCX-10, those that follow the NCX-10, and those that neither precede nor follow the NCX-10. Drawing rough-cut diagrams is fine with me.
4. What are the production-scheduling principles discussed in The Goal? (You should be able to list at least four or five such principles. As an example of such a principle: “an hour saved at the bottleneck is an hour saved for the whole system.” Both state the principle, and then explain its rationale.)
5. Provide an explanation of the pitfalls, as discussed in The Goal, of using cost accounting data for manufacturing decision-making. For example, why did it make sense for them to buy and extensively use the robots in the first place?
6. Would you guess the plant Alex Rogo manages is most like a job shop, a flow shop or a batch process? Which dimensions of competitiveness (price, availability, customization, innovation, and quality) would (you guess) Alex's plant is naturally well suited to provide? Which dimensions is not well suited to provide?
7. What are "dependent events"? What are "statistical fluctuations"? Use a simple example to illustrate how these two phenomena affect the throughput of a production process. In the scouts hike process described in the book, how do dependent events and statistical fluctuations manifest themselves? In other words, what is the “phenomenon” that Alex notices in the hike that can be explained by the above-mentioned factors?

8. What are ways/ suggested best practices we account and manage “unpredictable variability” in our supply processes, internal operations processes, and market demand processes for either manufacturing or service environments? List 2-3 ways for each one of the above three processes, and some examples you are able to provide on the way they are practiced and their effectiveness.
9. Why did Alex put Herbie at the head of the line? What are the implications for the desired location of the bottleneck operation in your process? What happens if that is not a feasible option for your process? How are you going to manage the process then for improved throughput and efficiency? In “lean theory”, where would we like the bottleneck to be? Why?
10. A frequently mentioned principle in the book is: "Do not focus on balancing capacities, focus on synchronizing the flow." Explain by using a simple example of why we should not be particularly concerned with balancing capacities in a production process, but be more concerned with balancing flow with the market.
11. Another frequently mentioned principle in the book is: "Resources must be utilized, not simply activated". Explain the logic of this principle by using as an example a situation described in the book. What are the reasons that often companies end up activating their resources instead of knowing how to effectively utilize them?
12. In Chapter 28 of the book, it is suggested that it makes sense to cut the batch sizes in half at the non-bottleneck machines. Do you agree? Explain why or why not. How do you know when you have reached the limits in cutting your batch sizes any further?
13. Why does their cost measurement system indicate that their cost per part goes up when they cut their batch sizes by half at the non-bottleneck operations (see Chapter 29)?
14. In Chapter 21, in an effort to improve the throughput of the system, they introduced a priority scheduling system using "red and green tags". What is the rationale behind such an action? Explain. In Chapter 25, they discovered some problems with the use of such a system. What was the cause of the problem? Explain the logic of the remedial action required to address it.

The above is an individual assignment. It should not take more than five or six double-spaced page(s) to answer the questions, but treat this as a “guideline” rather than as a page limit “constraint”. Use as much space as you need to clearly express our thoughts.